Wearable Smart Insole for early Diabetic Foot Ulcer Prediction using Multi-Modal Sensing

INFORMATION TECHNOLOGY UNIVERSITY

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Introduction

- Diabetic Foot Ulcers (DFUs) are a serious diabetes complication that can lead to infection or amputation if not detected early.
- This project introduces a smart system that monitors foot health in real time using AI and data analysis.
- It helps identify early warning signs and supports timely medical care to prevent complications.

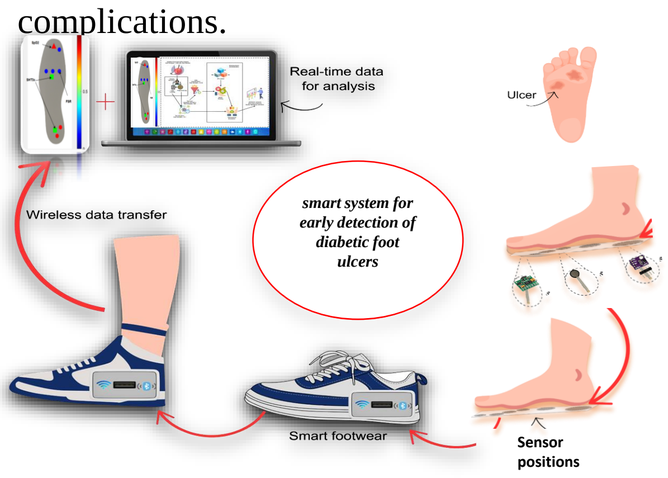


Fig. 1 Foot insole with integrated multi sensors system for complete foot real-time wireless monitoring of DFUs.

Objectives

- Develop a smart insole system that continuously monitors foot pressure, temperature, humidity, heart rate, and oxygen levels to enable early detection of Diabetic Foot Ulcers (DFUs).
- Analyze the collected sensor data using MATLAB and AI algorithms to identify early risk factors and patterns indicative of ulcer formation.

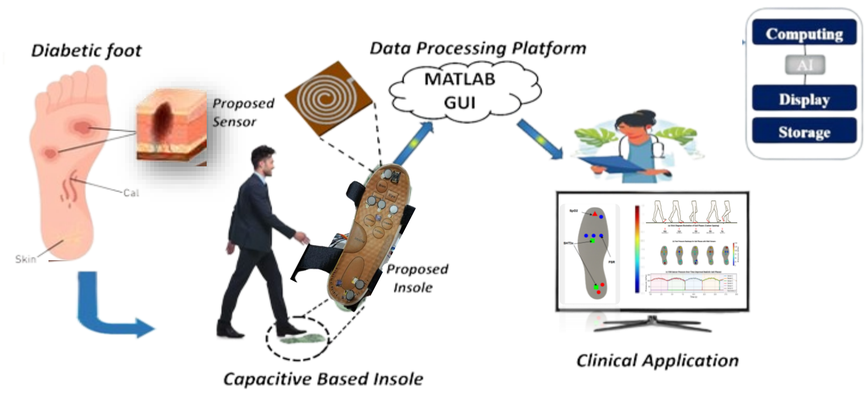


Fig. 2 Capacitive based insole, detects real-time plantar pressure distribution for DFUs monitoring.

Characteristics

- This system integrates multiple sensors within a smart insole to monitor foot health in real time.
- FSR sensors capture real-time pressure data from six key foot regions and continuously monitor SpO_2 , heart rate, while temperature and humidity tracking helps detect early foot issues under varying conditions.

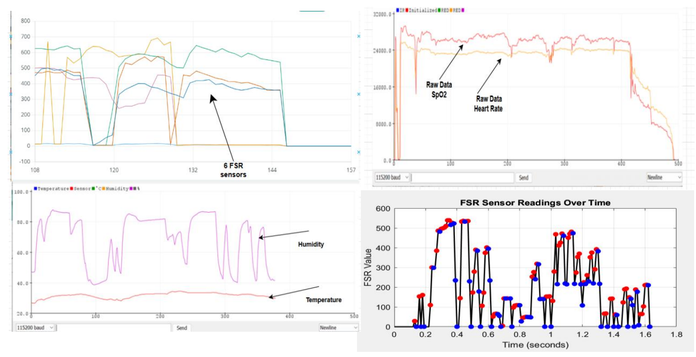


Fig. 3 Multi-sensor real-time monitoring for diabetic foot risk detection.

Methodology

Design Principle

- Capacitive pressure sensors were designed using COMSOL. These are integrated into a smart insole with multi-sensor capability. Sensors track pressure, temperature, humidity, and vital signs. Arduino board collects and processes the data in real time.
- Data is processed in real time using MATLAB and AI to detect early signs of diabetic foot ulcers.

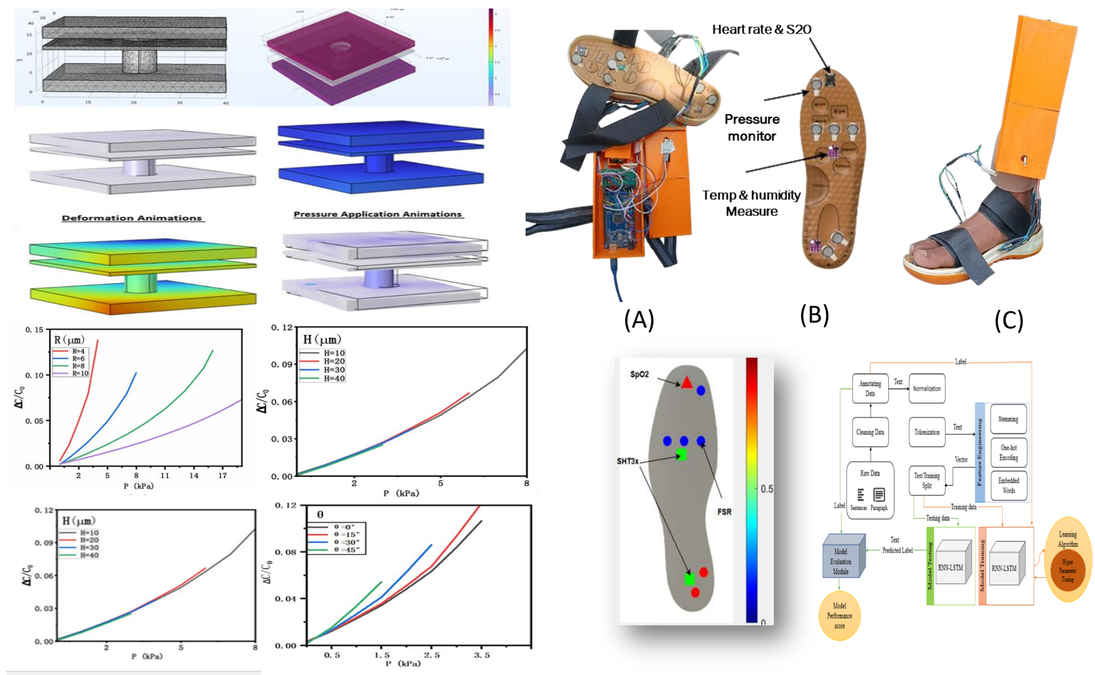


Fig. 4 sensor design, working, sensing principle and AI implement for classification & monitoring DFUs.

Key Findings/Results

- The system accurately monitored pressure, temperature, humidity, heart rate, and SpO_2 in real time. The Abnormal values were detected in high-risk foot regions, indicating early DFU signs and AI analysis enabled timely alerts and remote health monitoring.

Label	precision	recall	f1-score
Moderate/Infection	0.8	0.889	0.842
No Ulcer	1	0.885	0.938
Ulcer	0.977	0.98	0.982
accuracy	0.9538	0.953	0.9538
macro avg	0.9256	0.92	0.92117
weighted avg	0.95709	0.95	0.9544

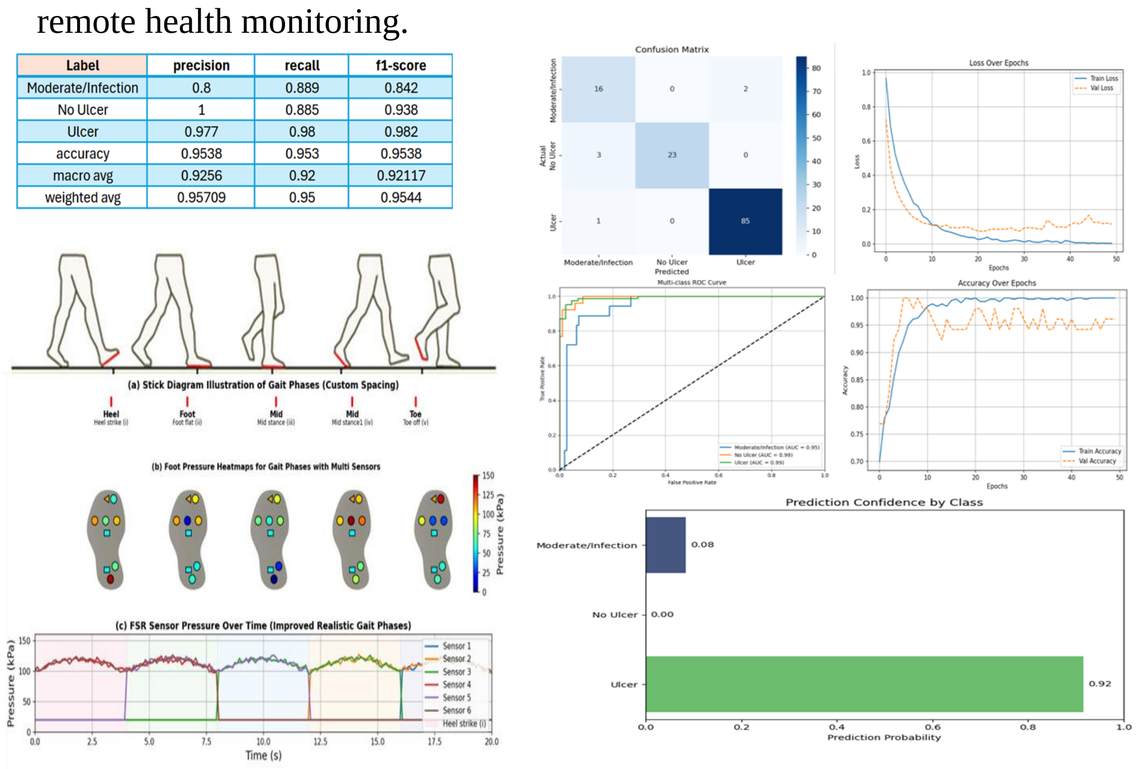


Fig. 5 AI-based gait analysis and real-time foot monitoring for early DFU detection..

Conclusion

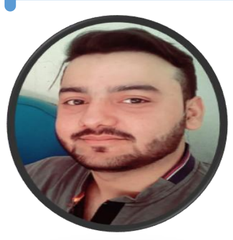
01 Smart footwear embedded with Low-cost and flexible IDC sensor-based insole.

02 Enabling a deeper understanding of how foot mechanics impact overall health in diabetic patients

03 Detects irregular and unbalanced pressure distribution that can lead to foot ulcers.

04 Provides a user-friendly interface designed for real-time diabetic foot ulcer monitoring.

Team & Acknowledgement



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SUSTAINABLE DEVELOPMENT GOALS

3 GOOD HEALTH AND WELL-BEING